

AAKASH SIRICILLA

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SUMMARY

Software Engineer with 3+ years of experience designing scalable systems and robust front-end solutions using JavaScript, Java, Spring MVC, HTML5, and CSS. Developed high-performance architectures that ensure cross-browser compatibility and effective performance tuning. Committed to agile collaboration, creative problem-solving, and delivering technical solutions that meet scalability and security standards.

EDUCATION

New Jersey Institute Of Technology

Master of Science, Computer Science

Sep 2023 - May 2025

Newark, NJ

- **Coursework:** Algorithms and Data Structures, Software Engineering, Machine Learning, Database Systems

Kommuri Pratap Reddy Institute Of Technology

Bachelor of Engineering, Computer Science

Aug 2019 - Jun 2023

Hyderabad, India

- **Coursework:** OOPS using C++, Operating Systems, Network Security, Distributed Systems, Computer Vision

WORK EXPERIENCE

New Jersey Institute of Technology

Research Assistant

Aug 2024 - May 2025

Newark, NJ, USA

- **Model Design & Optimization:** Developed LSTM and CNN architectures in PyTorch to detect SIP anomalies, raising accuracy from 78% to 93%. Tuned hyperparameters and implemented early stopping to boost generalization and reduce overfitting.
- **Data Pipeline Engineering:** Engineered scalable pipelines processing 120K+ SIP messages via Python and Pandas, cutting pre-processing time by 35% and doubling training throughput for experiments.
- **Performance Monitoring & Visualization:** Created real-time dashboards to track precision, recall, and F1 scores using Matplotlib and Seaborn, reducing debug and evaluation cycles by 30%.
- **Reproducibility & Collaboration:** Dockerized research environments to ensure consistent training across systems, reducing setup time by 70%. Led Agile-style stand-ups to align model iterations with publication milestones.

Cognier Insights

Software Engineer

May 2022 - Jul 2023

Hyderabad, India

- **Microservice Architecture & Scalability:** Spearheaded migration from monolithic architecture to AWS-hosted microservices (Docker, Kubernetes), reducing deployment time from 45 minutes to 5 minutes and improving system uptime to 99.9%.
- **Backend API Development:** Built and secured RESTful APIs for pricing and inventory modules using Spring Boot and PostgreSQL, enhancing reliability by 40% during high-traffic sales events.
- **Responsive Dashboard Development:** Developed responsive web dashboards using React integrated with Spring Boot, leveraging HTML, CSS, and JavaScript to ensure cross-browser compatibility and high performance for 10K+ daily users.
- **Security & Automation:** Engineered secure authentication with Spring Security and JWT, mitigating unauthorized access incidents by 90%. Introduced end-to-end CI/CD automation using Jenkins and GitHub Actions, accelerating deployment speed and boosting delivery cadence by 4x.
- **Quality Assurance & Agile Leadership:** Partnered with QA to design regression suites, reducing post-release bugs by 40%. Facilitated Agile ceremonies achieving 98% sprint completion and earned "Innovator Award" for process efficiency.

Across The Globe

Developer Intern

Sep 2021 - Mar 2022

Hyderabad, India

- **API Development & System Modularization:** Developed and optimized 5+ REST endpoints in Java and Spring Boot for authentication and reporting modules, laying foundation for a future microservice transition.
- **Database Optimization:** Refined PostgreSQL queries and indexes to reduce query execution times by 30% and boost response speed by 20%, improving user experience for the internal CRM system.
- **Testing & Version Control:** Increased unit and integration test coverage from 40% to 70% using JUnit, preventing production issues. Adopted Git branching and peer-review standards for code quality and traceability.
- **Agile Collaboration:** Contributed in daily scrums and sprint reviews to improve velocity and reduce backlog by 15%, gaining hands-on experience with professional SDLC practices.

PROJECTS

Turtleback Zoo Management System | Laravel, PHP, MySQL, JavaScript

- **Operational Automation:** Developed a role-based management system with JWT authentication, reducing manual admin tasks by 40%. Optimized MySQL queries to improve load times by 25% and boost user engagement by 30%.

Decentralized Milk Supply Chain Platform | Python, Django, Solidity, AWS

- **Blockchain Integration:** Built and deployed a secure Web3 solution for 200+ farmers, eliminating middlemen and ensuring transparent payments. Containerized Django and smart contract services on AWS for 99.9% uptime and 40% error reduction.

Real-Time House Price Forecasting Tool | Flask, scikit-learn, React

- **Predictive Modeling & Visualization:** Implemented Random Forest API ($R^2 = 0.90$) with sub-200 ms latency for 100K+ records. Integrated React frontend to visualize market insights, reducing estimation error by 20%.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, JavaScript, Typescript, C/C++
- **Backend & Databases:** Spring Boot, Flask, gRPC, Redis, Nginx, Kafka, MySQL, PostgreSQL, Elasticsearch, MongoDB
- **Web & Application Development:** HTML, CSS, React.js, Node.js, Express.js, Responsive Web Design, Front-end Engineering
- **ML/Data Engineering:** Scikit-learn, Pandas, NumPy, PyTorch, TensorFlow, Matplotlib, Seaborn
- **Cloud & DevOps:** AWS (EC2, RDS), Docker, Kubernetes, CI/CD pipelines, Terraform, Azure
- **Tools & Platforms:** Git, GitHub, Jenkins, Jira, Tableau, Postman, VS Code
- **Professional Skills:** Self-directed, Analytical Skills

CERTIFICATIONS

- AWS Certified Developer - Associate
- Microsoft Certified: Azure Fundamentals
- Secure Full-Stack Development & API Security (workshop)